

ActInf Livestream #028.1 ~ “Towards a computational phenomenology of mental action”

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Duration: 1:51:00 | **Uploaded:** Unknown **Transcript method:** auto_caption

hello welcome to the actin flab live stream number 28.1 it's september 7th 2021 welcome to the active inference lab we are a participatory online lab that is communicating learning and practicing applied active inference you can find us at the links on this page this is recorded in an archived live stream so please provide us with feedback so we can improve our work all backgrounds and perspectives are welcome here and we'll follow video etiquette for live streams today on september 7th and next tuesday we're going to be discussing this paper on computational phenomenology then we haven't set the paper for 29 yet and then we have a few papers set for october today the goal is just to learn and discuss this very cool paper towards a computational phenomenology of mental action modeling meta awareness and attentional control with the parametric active inference and we're really appreciative that lars the first author is here to discuss with us so we're just going to start wherever we start and go from there i guess i think each of us here probably brought some questions and anyone who has a question the live chat can ask it so let's just introduce ourselves um [Music] and end with a author and go from there so i'm daniel i'm a postdoc in california and i'll pass it to dean good morning i'm dean i'm retired i'm in calgary i'm in canada and the work that i do ties in nicely with this idea of being able to bring some precision to how we learn so i will say who can i pass it to so dave um david he's usually on mute yeah he might be on text-to-speech mode which is fine right but um welcome dave and we'll pass to lars so lars welcome to this discussion and just thanks again for joining we're happy to hear any introduction and then we'll go from there yeah sure well thanks excuse me thank you thanks for having me i'm kind of excited to be here it sounds like a fun couple hours ahead of us um uh just sending regards from the other co-authors as well obviously like a lot of these papers this has emerged in collaboration with uh other people and they couldn't make it here today so um but i'm glad i'm able to be here so my name is lars um i'm based at my affiliation is with the university of lyon in france i'm doing my phd there and my focus is explicitly on meditation actually and that's

what i'm interested in is what brought me into academia i had a first stint in academia in physics a while ago and kind of left it and went into corporate stuff for a little while and then it was drawn back in by i'm disinterested in meditation sorry there's a bit of background noise i don't know if you can hear it but um yeah i'll i guess i'll leave it at that the the kind of genesis here was from that from that angle through meditation and ending up in in carl's lab and asking questions around the tension and meta awareness um through that lens of of practice and that's where this kind of paper emerged from so we'll get into that a little bit more detail and where did active inference come into play was it from something you were learning about from that practice perspective like you mentioned or a theory driven perspective or did a collaborator bring it to the table it was a collaborator so i i i sort of came into computational neuroscience um looking explicitly to contribute to the field of meditation research and so i'd reached out to a lab in lyon one of the co-authors um and uh joined the lab there and they had a like an ongoing uh relationship with uh carl frisson's lab because jeremy matu is one of the co-authors here uh did his postdoc over at ucl and so they had a bit of an idea for an attention project um uh using active inference and i kind of came in not really knowing very much about it um but you get to learn and then i was sent off to london with a pretty open blank check as it were of like see if you can figure out how to apply this to meditation and attention and we had some some some directions that we were thinking about but then what ended up emerging was this uh hierarchical sort of deep parametric uh um stuff ooh interesting how it kind of came to yeah it really was right right right place right time they were looking for somebody the physics background with an interest in meditation and uh computational neuroscience and i sent them an email and that was pretty much the extent of the interview process and off i went cool well welcome steven um dean or steven if you have a question you can raise your hand or of course anyone in the chat um maybe dean did you uh want to talk about a crosswalk or something like that that yeah i want to do that later but i don't asking so it's fantastic to get the background about the meditation piece because when i before i picked up the paper what i was wondering about was how precision plays a role in learning and especially when you're moving from novel contexts and trying to familiarize yourself with something and so for me my my curiosity was around the idea that when you have the simple model and then you move to these um more nested paradigms you're kind of moving from the unplanned because you can't plan your perception to the plan the idea of being able to think about your thinking and so that transition was what really fascinated me so now that you brought up the idea of meditation i wonder how meditation goes from the unplanned to the

plan so you've opened up a pandora's box for me because now my original question is kind of now i've been taking down this tangent so i'm curious right well i mean i can speak to that a little a little bit um the the the the idea for me going in here was to see i mean just intuitively as soon as i got familiar with active inference there was a um a resonance there in terms of how the maths seems to echo a lot of kinds of ways that people talk about practice in general there's this sort of balance between um you know attention and the the that our perception is this this combination of prior and observation you know that idea and like how these two things are balanced and and getting into um the optimal space there is very very close to you know if you start thinking about mindfulness obviously is a way of paying attention and increasing the impact of the sensory observations but then also things like don't know mind beginner's mind uh this kind of fresh kind of uh stance that is very common in zen but in other in other areas as well and so already there's this kind of like oh there's something there's something here um and so i arrived at carl's lab and and what i started out with was just getting familiar with the attention literature because that's at least you know the a good way in for meditation because at least in the very beginning that's what it's all about and in some ways it's about that right to the end but um so the question was uh you know how do we uh talk about paying attention um within already and i was really fortunate because i was kind of coming into the space uh when all the scaffolding and the maths i needed to start talking about this in a more practice related way or a metacognitive way was already there all it really needed was somebody to come in and ask a question that was for me phenomenologically driven which was in meditation you are paying attention for sure right you are deliberately paying attention to some object you know traditionally the breath but really the aim of the game is uh or an additional part of that which is really really important is being able to monitor where your attention is so in other words being able to pay attention to attention and so that's where that kind of shift happened right you mentioned like thinking about thinking it's like well okay so if we know some of the maths about you know what it means to pay attention what would it mean to pay attention to attention and what would it mean to be able to have control over the intentional states and how would this how would we make that an allowable thing within within the framework um yeah so exactly the deep affective framework kind of set the set the scene for that um and casper casper tells work sort of gave the gave the hint and that was on a different precision in the model and here we're just talking about likelihood precision and then that kind of created this this um opportunity then right that then becomes almost generalizable across and that's and we can get to it a bit later but really what i'm more excited about in this in

this uh whole work and paper really is what we start to hint at at the in the discussion at the end um pointing to the fact that we could generalize this across precision throughout the model that could then be interpreted in as different forms of mental action and what that might mean for understanding yeah i mean it gets broad at that point yeah awesome thank you for the summary and welcome blue at all at feline so stephen with a raised hand and then any live chat or anyone else with a raised hand yeah welcome um yeah i was i'm really curious i like the what you're saying about the generalizability or the potential for generalizability and you're sort of extending beyond neuro phenomenology to phenomenology in general i suppose is a name i'm curious how how you see action potentially coming in with meditation because you talk about sort of reflecting on mental states like breathing is an action i suppose but in many cases they use meditation because you're reducing the amount of action that's happening in the amount of visual stimulus because of often the eyes are shut you know bodies contained and um and now i see that you know okay so how does this then come back into an action state and i know that there's a kind of in between with like i know vipisana type meditation on the body senses or like you mentioned the breath but i'm curious how you see that kind of inactive piece starting to come in through this no thanks for the question and it's a really central one um because i mean i guess to give a little bit more history to the this paper there was this is also building on a little bit of some of the intuitions of previous work by antoine jeremy and another collaborator giuseppe pagoni um and they wrote a paper called on the epistemic value of inaction uh which speaks to this this point um i think in the begin off from the from the outside we can kind of separate you know mental action and over action and meditation you know you're putting yourself in this situation where you you know deliberately don't have that much over action right you're sitting still um and yet you're doing things you're definitely doing things with your mind um even sort of do nothing meditation is still a kind of doing in a sense um although it gets a little bit sticky um but the for me to answer your question the cessation of overt action puts you in this highly epistemic state because now the sort of self-evidencing side of the equation you're putting yourself in a situation where that is harder to access so that you you no longer can fulfill your expectations through you know action acting in the world right so like um if you were to sit down and start getting um agitated or bored rather than the sort of fulfilling the expectation there that you shouldn't be bored by going and doing something you have to stay stay bored and there's a certain discomfort that can arise from that and then the role of mindfulness or acceptance in that moment it becomes central but then what it allows then for is that that your model has to update right if the model can't be

sort of you can't find evidence for it well then you gotta you gotta update it and there you get to this notion of vipassana right which is like inside clear seeing inside right is that you're learning something through through that that wasn't available to you the moment the moment before and like apologies to everybody out there who's like maybe very into the meditation that like this is very sort of hand-wavy way i'm talking about it especially kind of like matching it over to the over to active imprints a very sort of loose way but that was kind of the the gut um intuition anyway coming into it and remains remains the the intuition and i think there's this really for me anyway and where i'm kind of seeing my my career with this going just is the very beginning of an exploration of how to how to articulate meditations different practices what they how they might cash out in terms of um precision dynamics different forms of learning different forms of optimization and what that might mean you know for us so i think that's why i wanted to start with attention because it's pretty uncontroversial as it were and just start building up the building blocks and be able to start making more and more interesting inferences about what it might mean to be doing other kinds of of practice awesome answer thank you blue or yes go for it blue hi so i don't know if you had a chance to watch the discussion or like the preliminary um discussion we did yesterday but something that came up was in the idea so daniel can you flip to the circle about like uh tension and here the cycle this one yeah that one so something that came up there uh when we were discussing this figure was like the the discrete steps between like distracted awareness of distraction redirection of attention and focus like you know there's kind of like an intuitive um feeling that the the distracted awareness of distraction and redirection of attention back to focus is like discrete but like there's a transition that's kind of muddy going from focused to distracted and something that i thought about was like you can become so intensely focused on something and like the example that i'll give you because you'll relate is like going down a scientific rabbit hole like we're chasing reference after reference after reference and so you're still on task and you're still correctly paying attention to what it is you're supposed to be paying attention to but like you're so focused that you're finally off task right because you just start getting interested in other things and so it's kind of a gradual transition from focus like you're still focused but you're you're like focused on the wrong thing and so i just wasn't being like where in this model do you think that kind of like focus until you're so focused you're not paying attention like where does that like come into this or how would you think to model something like that yeah thanks for the question so this uh model that we're seeing here is of course a bit of a caricature right it's it's it's a it's an intuitive sort of description of what's

happening because you're right there's degrees of distraction and you know meditation we talk about you you actually can classify these you know those when you sit down to try and pay attention to the breath there are times when you've totally forgotten that you were even meant to be meditating and you were like gone right you're thinking about something completely different and then the timer goes off and you become aware that you were completely gone but then there's other things that are much more subtle right this like maybe you notice some sound in the room or you know some sensations in the body but still somewhere in your awareness the breath it remains present but maybe less and more in a background foreground kind of situation and so like are you distracted then or not and to what extent um and all these things and kind of everything in between right um so in terms of how that would cash out here the first thing to kind of point out to is that the the precision parameter that we're talking about is a continuous parameter and so it doesn't have to be um relating to discrete states and we could have been free to um grand euro like make more granular the attentional state so it's not just distracted or or focused um and sort of have grades of of focus would have been possible but then the other thing that isn't modeled here is what are you distracted by um here distraction was just uh not paying attention to the thing but in reality distraction is your attention being grabbed by some other stimulus um and for some particular reason that is not uh shown in in this way that's something that we're hoping to start to work on in the in the future part of the reason why uh we didn't go down that direction here in detail is because there's still a i think a really central part that is is difficult to tease out still at the moment which is attentional resource you know in in some ways you could you could run this simulation and given the preferences on wanting to be paying attention if you put that preference on a bunch of different observations there's nothing stopping the agent just paying attention to all of it right maximally and having really high precision observations across the board now that doesn't seem to be what we're able to be there seems to be like a a finite um something finite about it and so when you're paying attention to one thing it's at the exclusion of something else and maybe part of practice and part of you know what you're getting in meditation is that that you know is softening a little bit your ability to sort of pay attention to more than one thing or have more allocation resources there in some way that might be learned over time but that's not um that's not shown shown here which is why you don't get that what you're talking about that kind of intuitive organic uh feeling of what it means to be what it really means to be distracted awesome i have a question and then blue so we've talked a lot about how spatial goals are set and even how some symbolic goals are set like sentence

structure but how are cognitive goals set so are they reducible to sensory preferences if so just the preference for you know some salt and sugar that's not going to really reduce your options in the city so how do we set those higher level and these increasingly like abstract and ephemeral cognitive goals yeah no thank you for that daniel um this is the the question i'm working on right now actually and on a follow-up paper where i'm trying to take this model and and specifically model uh focused attention in that and in practice in general the the question of motivation and goal is really central and where where in the model are you actually putting like how are you driving the attention behavior or meditation behavior one of the ways the way that i did it here was to say that in this kind of setup that the agent is given an instruction hey i want you to pay attention to this stimuli and that that then translates to them wanting to see themselves um in an in the attentional focused state and so the way that the dynamics here are driven is by having a preference on that attentional state um and observations of that state so the kind of like metacognitive observations of seeing yourself paying paying here yeah thank you so oh there um we put a preference on uh those ones exactly um and uh just as a side note there you might think like oh but do those observations uh exist really are they sincere observations what kinds of observations are these um and i think that's a valid valid question and something we talked about for a while but intuition here is that you know if i were to ask you the question um what are you paying attention to right now right and you're gonna be able to give me an answer from having observed something um and so there's some kind of an observation there that you're able to make that you can use to infer uh what attentional state that you're in so anyway that's where the preferences is is uh you know i prefer to observe myself exactly so then the question is what is it that drives you know like a practice and i've been inspired by some of the work by uh mirza and colleagues on sort of scene classification or scene as a foreseen perception we're there there seems to be this attentional driving of attentional policies given another kind of preference which is to resolve a scene which is to clearly perceive a scene um and that's actually the way that i'm i'm i'm starting to think about um at least in a practice context in like a meditation context because in reality when you're meditating it's not so much that you're trying to pay attention to the breath although we talk about it like that it's more like you're trying to be continuously aware of the breath um so you have some kind of a preference on perception or being able to be sort of being able to resolve um a hidden state and that that is then what drives the kind of dynamics behind the scenes that's pretty interesting thank you for the response blue sure that kind of slides in nicely to um what i was gonna ask so as a follow-up to my question earlier and also

to daniel's question daniel can you flip to the mind wandering side yes thank you i had a question here um and these are two different models the one on the left specifically is from the mind life institute by a phd who works with um you know meditators right and so and then the one on the right i'm not sure if that's an explicit uh meditation task but i just thought there was an interesting context here so this paper on the right which is the neural model of mind wandering so you have like this paradigm where you're on task which is like in the meditation context say focus on the breath or i mean maybe it's not focused on the breath but i just i think about meditation as you know being present in the in the present right being in your in your body aware of your hands feet breath etc yes so there's this model where you're on task which is like focused on what you should be focused on in present and and then mind wandering they talk about as the pursuit of internal goals which i thought was interesting but i think about that in terms of like thinking about your shopping list or all the things that you're not supposed to be thinking about while you're meditating right like just going off and pursuing your own internal goal oriented thoughts and then they have a separate like category for being actually off task which is like where you're not thinking about anything you should be thinking about was just thinking about like random like whatever nonsense um and so i just wonder like do you or could you think about making those kinds of distinctions in this kind of like and it goes from from being focused on the right thing to focus on the wrong thing like where does this off task come into play i'm just curious if you have any thoughts on that i'm not exactly sure how they talk about off task but what i could just talk about here is the distinction between to me the the distinction here comes down to the meta awareness piece right where on task is pretty understandable off task you could imagine just doing something else but aware of that fact you know like i've just decided to change and start focusing on something else but there's sort of a metacognitive awareness of the fact that you are doing something else whereas mind wondering is this uh slightly or at least the way that we're talking about it here um different situation where you're distracted or you're paying attention by when i said distracted just been paying attention to something you hadn't intended to pay attention to but also not aware of that fact right you haven't updated the fact that you're that that that's happening um and that's the crucial piece here that we're trying to regulate right with uh practice or or whatever in in an intentional in an attentional task for that distinction i think i think um active inference kind of you know it almost gives you that for free in a sense to kind of show you where where that phase would be and that's what we were kind of showing in the results is that that period of of updating of where your attentional state when you sort of think it's one but it's actually another

um so you think you're paying attention or you're somewhere you don't really know where you what you're doing but you're pa you're distracted is then how we would sort of express that or cash that out computationally blue is that a quick follow-up yes in terms of active inference um so it is so i just wonder is like off task or am i wondering like i had never thought to split those two apart until i saw this this paper and maybe check it out if you have time at some time but but really like i think about this mind wandering is this like epistemic exploration is is that like i mean is it exploitation when you're on task and you're focused on what you should be and then is it you know we think about active inference in terms of optimizing exploration and exploitation uh is there a room and a model for that epistemic exploration or is it always like a bad thing is it always a bad thing you know i don't have a i don't have a good answer for you to be honest this is what we kind of want to go towards when i was saying you know what are you distracted by and what's the why behind you know the default bone network basically like what is what is the value of that on the one hand i could imagine that yeah there's a there's sort of like an epistemic wandering through state space and future possibilities and just sort of monitoring of potential negative outcomes in the in the future um with for which there's obviously value um and i don't think it's necessarily a bad thing i think when when i you know in a practice context anyway it's like that that action can also lead to discomfort now and anxiety now um because you're sort of simulating future bad events and there's actually a great paper casper wrote a good paper on that um deep i've read it yeah yeah the simulated future events well and so often meditating like frees up space in your mind i mean i don't know if you've had the experience but like where you're not thinking about some other problem but like you'll be meditating and like whoa have this like brilliant flash of brilliance or like key moments of insight right like be possible so there's just like ins but it's maybe not insight into like some internal like deep spiritual discovery but it might be like insight into some external problem that you're having so there's there's that and so i wonder like that might be the difference between off task and mind wandering like if you're off task it's like maybe a non-beneficial or like there's no excuse for it's just like doing blah blah whatever and then mind wandering might be like exploring your neural space and in like a recombination phase and think about that so anyway it's just something to think about going forward interesting interesting i think the the the notion of insight kind of why practice leads to that is really exciting place to start thinking about like what is it about paying attention in a particular way that leads to these moments of aha thanks i'm going to read a question from dave dave wrote a practice i've been following based in naropa institute's shamatha practice but not much stress

there is to attend only intermittently say permit mind wandering during the out breath but attend during the in-breath there are two effects obsessional thoughts are disattended without intending to disattend to any content the motion of this attention dis of the transition from disattentive to attentive occurs many times the usual return gently to resting the attention on the breath happens several times a minute rather than several times per hour does anyone else do this or discuss it maybe to warn against it close parentheses hyphen to your knowledge i like that i didn't know i haven't heard of a practice like that before but in the way that you're describing it i could see a way that would be really really powerful and really useful because one of the pitfalls i can tell how this conversation is going to go more and more into but but i'm i'm i'm i'm up for that i'm game um but one of the one of the pitfalls of course in in in practice is is all the subtle ways in which resistance can come up towards our own minds and if we are paying attention and we start to see distractions as things that are stopping us from paying attention and that we don't want right then it creates this tension and is actually counter to what we want to be going for and i could see how by deliberately letting mind wondering happening and coming back it sort of cuts that tendency to have resistance to to thoughts which actually just ends up perpetuating them or giving them um more uh potency in the mind whereas really the aim is to pay attention to the breath and it's funny because when we say pay attention we think like it's this like focus you know thing in reality 90 of it is relaxation and letting go and letting go of resistance to distractions distractions come and then we come back back to the breath and that's what kind of does does the work so i would love to um i'm gonna look into that a little bit more detail that style of style of practice would be interesting to see when we have a more fleshed out version of this like how um different approaches to practice might lead to different kind of learning outcomes at different rates etc that topic of distraction it's like if somebody were a captain and they cited something very subtle that wouldn't be a distraction that'd be on task so then there's this aspect of the multiple levels like there's the eye circading and then there's higher higher and modeled in this paper greeting scott at slower and slower kind of clicks of the model and uh so it's just interesting yeah what a distraction implicitly says that it's not on task because if it was even something that was unexpected that drew the regime of attention that way if it were relevant for the task then it would still be considered not a distraction part of the relevant stimuli for the task but the exact same stimulus might be off task in a different context like five minutes later yeah yeah completely i mean if somebody's trying to talk to you and you know share something with you you only paying attention to the sensations in your hands is not very helpful um but i think what we're kind of

flirting with here as well is is this notion of task and goal and on task and off task i think in a i mean in a meditation context that's a you know um difficult difficult thing to define there's a big difference between somebody who comes into practice because they want to you know sleep a little bit better or not be so stressed or have a little bit you know more ease in a relationship to somebody who comes in and says you know i'm in it for really understanding myself and who i am and what i am and for you know questions deeper questions of like awakening and um you know expanding awareness and these kinds of things and those intentions and motivations are you know these task priors that you're setting over this whole thing and are driving the way that you practice and you know traditionally what is sort of said again and again and again is that defining that is pretty important in terms of what you're going to get out of the practice because to pay attention to the breath to just pay attention to the breath without really understanding why and where it's leading to leading is um is one thing whereas having it within a context of you know a wider program and goal is is a completely other other situation now modeling those in active inference is interesting right i would say that the the latter sorry about the background noise here but the latter example there of you know thinking a bit more broadly is is having some kind of a your own happiness right like your own model fitness ultimately you know if we kind of go on the affected inference kind of view of you know casper tell that like affect in some ways related to how well you're doing and how and how well your model is is predicting and so to have that as the as the intention rather than just paying attention to the breath but seeing how paying attention to the breath can lead to that ultimately is a different kind of goal goal statement than than the sort of narrow attentional task itself thank you lars dean and then scott yeah i like to think you're i think like no i'm on okay i think you're taught i think your topic lends itself to wanting to gallop ahead because it gets people so excited about this idea of moving from subliminal to liminal to kind of a limital like a precision around edges right so that transition gets people excited because now they say oh i can think of a ways that i can use this in whatever it is that i'm trying to figure out um so at some point that transition pulling that transition apart but not breaking it is is going to have to we're going to have to sit down and really parse that with you because it feels to me like we're doing a a is are we 28 or 29 we're doing the 0.2 right now because everybody wants to really get this thing going because it because it's uh it's kind of exciting stuff right like you've given us this this backpack of stuff and we're like opening it up and it's all this candy and we're but i'm still really kind of interested how that how that goes from subliminal which cannot be planned and cannot be thought out because we're talking

about act sensory inactivation thinking about thinking because when i when i see the diagrams themselves they're there but it doesn't really help me understand that any better because i still think that's a really really tough bridge to build yeah you know what i'm saying i know it's there but it's still hard for me to get my head around i think i think i know what you mean being what you're referring to which is like where does the where does the structure come from in the first place is everything yeah well we know where it comes from in the first place and we know where it ends up it's the transition in between it's those casper got us to the third layer now we've gone to a deep transitional we generated a context we're thinking about our thinking it's metacognition it just appeared it just emerged so how does the con how does the math work on that i i think i think what you're getting at there i mean there's kind of two ways that you could think about a model like this is like either you say well the structure's in place and what's changing through learning is the precision on observations essentially like you become aware of it it becomes something that can factor into your perceptual uh belief updating um that's one way around or you can say that there's something about human development and maybe the evolutionary process that sort of leads to this or structure growth and learning um and then the question is well how does that come about the first question is a little bit easier to answer than the second question because obviously structured learning with an active inference is still an ongoing area of you know investigation research and and how exactly that comes about questions of neuroplasticity and things is quite it's quite difficult to grapple with and then how why do particular practices and you know different paths in life lead to certain things than others but then to take the other stance or the other way to talk about this which is i think what you were saying correct me if i'm wrong but like let's assume that the structure is here that we have some kind of a deep hierarchical um you know parametric structure that is giving rise to to metacognition how do we go from being uh not aware at all of the contents of our mind to being aware of the the contents of our mind right yeah it's kind of contradiction right because the way it's set up is it's a bridge to everywhere it's not a bridge to somewhere and then somehow in that bridge to everywhere metaphor we select out a particular and we give it our focus or our or our attention and so i'm not saying that the that the math doesn't work i think the anchor piece is that at the basic level nobody's going to argue that we have a markov blanket and we're going to we're going to statistically move closer to what we think is going to happen next we're going to update it's the it's the last piece which is we're now thinking about our thinking that just seems to pop out and make itself available you know because we've now built up a set of priors but we i don't i i'd like to see

how the math explains that that's what i'm because it's still it's still a distributed set until it collapses to what we focus or attend on and so it seems like a contradiction and maybe that's what it is maybe that's what the math is trying to explain to us i'm still not sure if i've completely uh grasped the the point of interest here what what what all that how the transition works again i'm sure you'll be able to explain it but what i'm seeing is a transition first starting up being very open and then the mind is able to buckle down for lack of a better expression and get its attention on something it starts out open and then it moves to something closer what i'd like to see is how the math explains that that make sense i think so i think so dean maybe one way just saying it would be just looking at figure five like the lower levels are certainly outside of our control and the higher levels are moving towards phenomenology whereas the lower level in traditional neuroscience would be seen more as like sensory processing and yeah it's indeed similar kinds of models so how do we kind of bridge not just the um science humanities gap but like the parameter experience gap and then what is that sort of marsh between them and is it really the case that just by nesting this model that we know and love we ended up getting the or a structure of cognition is that the or a structure does that exist as part of a broader family of related models is it just about finding patterns of nodes that explain variance in multiple components of phenomenology and physiology that's the question well i for me it all comes down to precision uh within the model uh then that's kind of how i been thinking about this is that you know at the sensory level that first level are multiple places that you can start to identify just patterns you know statistical patterns and then start to layer on top of that a confidence or precision over that which you know we have evidence for you know how that's mediated in the brain for various various places but the precision then itself itself becomes this thing that you can layer on top of your entire sensory experience and gives you this kind of internal flexibility as to how you're how you're interacting interacting with the world and then the modeling of how those precisions are being driven you know without your awareness is then what what those states that that be the b matrix at the second level would be would be encoding is you're starting to learn ah okay so this you know my attention gets grabbed by this or that or like these are the kinds of and here we're talking explicitly about attention but there's other precisions around that you can start to understand how they are evolving and then also learning where you're where your reach of control is like how much of how much of this can i control and and influence and what does that mean and what are the impacts of that it's kind of what i find exciting about exciting about this because you can kind of see how different forms of precision control and awareness would lead to

different kinds of model optimization and calculate the free energy you know based on on how different precision profiles are defined and evolving and kind of over time that's something that you would hope to be able to learn and then the question is how do you actually learn that well by paying attention by paying attention to how it's changing um and that's not always easy um and sometimes things in life you know make us pay attention to it you know start to notice like i'm the kind of person who gets yeah thank you exactly this figure here that gets impacted in these ways but i think there's a really important thing to say around the role of a teacher in this as well somebody who can point these things out in doing so direct your attention in a way that you wouldn't have thought to previously into noticing like hey there's this pattern in my mind that i hadn't thought of as a pattern i just thought of as well i hadn't even thought of it it's just happening and then as soon as i start to see it as a pattern it's something i can start to model and then you get this kind of emergence of a of a higher level structuring or understanding or grasping or modeling expectation over these patterns that were previously unconscious thanks i'm gonna just read a comment with uh no response needed from the live chat and then go to scott so joshua benjamin wrote all psychic ability seems to be subconsciously driven oneness of the active and subconscious mind would be magical to say the least people that already have their active and subconscious mind working as one also seem to have identical palmistry between their left and right palms interesting observation scott yeah thank you i'm sorry i was delayed in coming in but i'm always thrilled as soon as i drop into the conversation it's like being with family so this is so wonderful what a great conversation i wonder if you could comment a little bit the attention versus intention it feels like part of what we're talking about now let me let me comment on that a little bit and it's following up on dean's um exploration there i was thinking i always talk to people about the aha moment if you know the if you know the um punchline to a joke and someone tells you a joke it's just not funny because you know the punch line and so the aha moment is the same as that moment of eureka when you get out of the bathtub and run around yelling eureka but you do you understand something it's a paradigm shift like thomas kuhn in the structure of scientific revolutions talks about that you shift from not knowing something to all of a sudden knowing it so those i'm an addict for those moments i love ignorance as a result so i wonder if you could talk a little bit about going from attention to intention i am coming to the belief from working with all you folks and others in similar explorations that the mind actually doesn't exist at all in the brain the mind is always in society and in language always the brain gets tuned into a local mind by exposure by and then that become the attention turns into

intention you adopt like a bauer bird you adopt nascent models from the externality there's no onboard feral consciousness in my assertion here so you develop it and cultivate it based on your exposure to the environment as you build it up the attentions your teachers out there in the world whether they're formal teachers or informal can help you convert attention to intention because the models get developed based on their their explanatory power of your environment right if they they get cast aside if they're not as helpful in closing free energy gaps for instance i just wanted to throw those things out there and following up with dean to me the paradox of the frizzle between consciousness and unconsciousness is not something that we can resolve it's something that we can observe and maybe manage in some ways with these models but the paradox is one thing i always told my kids is the only time you see reality is when you're seeing paradox and anytime you're seeing the absence of paradise you are seeing a and when people say they want to resolve paradox it means they want their model to win here it feels like as dan was alluding to that there are multiple models that are in play simultaneously and the multiple models maybe can give us a sense of the shift from attention to intention anyway your comments are welcome thanks scott thank you for that lots of interesting points there i'll just pick out one of the things you said there around insights and these aha moments and from and this isn't present here again in this work encounter falling into a dean's trap here a little bit of um running forward with it but if you imagine here that what what's being afforded by you know paying attention new ways the dynamics and and precisions you know within our generative model and how and how the all these things are evolving you know we're gaining in model kind of structure and complexity ultimately bits of work that i find really interesting from carl and other colleagues is around you know insight and they've been talking about it in the context of psychedelics specifically as these moments of model reduction um invasion model reduction whereby you lose a bit of model structure and there's this liberating aspect of that and there's this sort of knowing aspect of that too and it's not that you've sort of learned something new you've kind of unlearned something and in that you you gain in model fitness as well you know there's this kind of um improvement of the state of affairs by by doing that and so you could imagine a scenario whereby um in playing this game of paying attention to how things go and how things are in the mind specifically your model structure's kind of gaining complexity but you're also identifying places that can be reduced and you get this sort of cyclical growing of complexity and reduction which matches on really really nicely and i have to be really careful sometimes to not try and like recapitulate things that i um like from sort of practice theory into science and

really be you know grounded in in the active inference but nevertheless there's you know my heart kind of wants to say that kind of nicely matches to the cycles of practice too these cycles of of insight growth that does come in this sort of non-non-linear and cyclical way thanks lars so this isn't really like a question but it's more of a comment so scott was kind of talking about converting attention or intention to attention but i think that that's also something that's cyclical so you can convert attention to intention and intention to attention you intend to pay attention attention gives you an intention um sorry that's like way tongue twister but like where does imagination come into play there right like there's attention intention but where's the exploration the creativity and does active inference allow for maybe this component on attention intention and imagination cool question um i think of imagination a lot in terms of uh generative practices so like state generation practices like loving kindness gratitude forgiveness these kinds of things and for me the way that i think about that is you are using your imagination to generate goal observation the goal state right which then allows you then to set a prior over that to aim for so if you evoke in your mind the a you know may you be happy so i'm imagining a situation where you are happy and now that is a possible future state that i can put prior over once i've done that then the machinery of active inference starts you know coming into play to achieve that kind of goal and that's a very very hand waving very loose and again like i can i can sense that people out there might be cringing to this kind of way of talking about about these models but you know an answer to the question that's kind of how i how i think about at least that form form of imagination that's some point two stuff thinking about how our cultural environment scaffolds our preferences and our telios our end directedness which is something that came up in navigation now it's coming up in cognitive navigation and setting of these higher order parameters scott i just wonder why is it cringy from the math perspective the right because i'm a humanities guy and i'm the math always freaks me out so i want to understand though what is it about the you're a math person steeped in the math what is it about what you just said that makes you uncomfortable with the math and how might the math be why is it stretching the math or how might the math be extended to to accommodate that cringiness problem yeah no thanks scott that's a good question um the reason i say that is because these models can pretty much model anything right you can you can adapt them to make almost any point that you want um and so you know if i start saying oh this is how it is and this is what imagination is and does well then i can sort of ad hoc um play with the model until it until it until it makes that point um and what i've really tried to do here in this paper is be very careful with that and to allow the dynamics to emerge intrinsically

from moves in the mathematics that are like allowed and and and don't make many assumptions because really all we're doing here is having a higher level that is parameterizing precision on the lower level and then we're just running it and seeing what kind of dynamics emerge and interestingly the dynamics that emerge are aligned with the expectations but if i go too far the other way which is really easy to do with this right because you have especially in a practice meditation context there's really detailed phenomenological models of all kinds of uh mental dynamics and it'd be very easy to say you know just gerrymander active inference to describe all of that and so that's why i'm trying to just be a little bit careful on how i talk about it one just to follow up the one of the things this may be uh rationalization but so i was i'm a lawyer so i was a rhetorician and so when we have incommensurables in the law we use rhetoric to glue them together and it's not crazy glue crazy glue doesn't fill gaps its rhetoric is epoxy right it glues stuff together and fills the gaps right crazy glue just sticks stuff together and so the reason i'm talking about that is is it again i asserted before that the mind actually exists in language that's very weird and loose language i'm using there the mind exists in language it exists in material culture we're born feral and we get our brain gets tuned to a local mind let's go with that for a second as a posit so you you have nothing on board when you're born would is it i guess another way of asking the question can the math construct consciousness from a feral mind with no social exposure that's my question and i am beginning to believe that nothing yes there's imagination of course there's a lot of lateral thinking but all the things are borrowings from the externality is my assertion now that's not the imagination you can mix and match them different ways but if you're isolated and have no question is i guess will the math generate a consciousness without any external input and i think the answer is no and so therefore it feels to me like the brain is a mind instead instance and the mind actually doesn't it's paradigmatically weird and different i don't maybe there are people who say this i don't know i just don't know enough about the mind studies but it feels like we can allow ourselves to not feel cringy about the extension the math now again i'm not the math guy i'm the cringy on the cringy side of this but by if we recognize that the source of the mind stuff is external maybe that's okay and it's processed in the brain anyway i don't know if that's something he's responding to but it's to me this is very interesting because this is that metaphysics jump and is it metaphysical or not i don't know i i the i mean i i don't i don't uh have much resistance to what what you're saying there i don't think these things happen in a vacuum and i think um you know just intuitively uh you know the that kind of a cognitive structure would not emerge without some kind of you know

societal or back and forth multiple nodes in the network you know working working together on this and you know anecdotally a line that always sticks with me and um from the buddha of all people is uh you know there's two things that are required for awakening there's wise attention and the voice of another and it kind of speaks to that need for interaction now the reason the reason i the reason i'm what i really meant when i'm saying cringy is that i think there's actually a huge advantage in just going slowly and methodol like and systematically through this because what it allows us to do is be very very precise in what we mean with our words because i can talk about attention and that might mean several things to different people but it almost doesn't matter what the definition is of it i can just point here in and say well whatever this is what i mean by this word and i can keep doing that building up the complexity of it and i can start to have very detailed descriptions and articulations of mental dynamics that don't rely on interpretation of the words because they are referring to something quite um specific thanks um and it is really important to have that uh perhaps carl firsten would say a deflationary view so the deflationary view of this figure 12 the generalized deep generative model of mental action it's like i mean big if true the deflationary view would be that just like a linear model like a linear regression bayesian modeling can be used on any kind of data so just because there's a word that you may or may not have familiarity in front of it like frequented statistics bayesian statistics it doesn't get around the fundamental problems of data modeling and it actually has a lot of similarities with hierarchical gaussian models of control which are common at least related to this idea of nesting slower levels of policy inference with again tricks that are really interesting like free energy calculation all those other things that we talk about other times but that kind of model never took on a philosophical or metaphysical baggage even when it did appear to correlate or even have predictive power for example predicting what somebody would buy and um it just it's a choice that we make what is our cultural priors what are our personal preferences over connecting that metaphysical jump that scott mentioned maybe some people they um think it's a six-inch jump and they can jump 20 feet and other people they have a belief that it's a bigger jump so that's why it's such an interesting area to talk about because there is some hyperinflation going on and then there's some extreme deflation and there's everything in between as well yeah and if i could just uh yeah yeah larger than scott um the thank you yeah for for articulating that so well because um i know i'm also guilty of this in my own mind when i'm thinking about this privately of just sort of running running this forward and losing track of of of what uh these sort of ontological commitments in in this whole discussion and i think uh maxwell one of the co-authors he

sometimes refers to you know hard for estonians and soft fristonians and that kind of speaks this deflationary and inflationary inflationary point but for me in this in this work um what i'm most excited about isn't so much a an ontological proof or conclusion of hey this is what it is and this is what the mind is doing this or or anything that's strong what i'm more excited about when i think is more relevant to our times is that even if this is a a facility or a a kind of uh loose description or a what am i trying to say approximated framework of something that's going on if it allows us to articulate intelligently like how these mental dynamics are going on it really helps you know because i teach meditation as well and one of the big obstacles that you come up against is like yeah but why should i do this like how does it work and what what what's the point and if you have something like this that allows you to articulate why and how and the dynamics and what's going on then that breaks down that barrier and allows for in a conversation that's at least grounded in something closer to you know biologically inspired computational you know neuroscience rather than a phenomenological report not that there's space for that but for a lot of people this is what would get them through the door thanks lars scott and then anyone else with a raised hand so i just wanted to go back that's this is fascinating and i wanted to go back just that rule the idea of rules of general application is this reducible to it is our elements of this that are interesting reducible to formulas is what i'm where i'm going with this query now and i wanted and it made me think about you have this society you have different societies with different standards different music scales different recipes of food and things that people are familiar with right they're raised in that context so chinese food is and music are familiar to a chinese person if you're born in a different culture you don't those are unfamiliar to you so i wanted to talk a little bit about rules of general application because in law so i come from law and law you know we've got technical specs now and here we got fristen analysis and trying to find general rules of general application that's fine that's good stuff and they and the one of the things that's interesting about rule of general application is how general they are so in in law you have this thing called equity which corrects the rules of general application i just want to read you this aristotle quote that guides me a lot here aristotle said the source of the difficulty is that equity though just is not legal justice but a rectification of legal justice the reason for this is that law is always a general statement yet there are cases which it is not possible to cover in a general statement hence while the equitable is just and is superior to one sort of justice it is not superior to absolute justice but only to the error due to its absolute statement this is the essential nature of the equitable it is a rectification of law where law is defective because of its generality the reason

i'm raising that here is you know we can find general rules they don't have to work in every edge case right and if we find edge cases that undermine the general rule it doesn't mean the general rule is no good right it just means we need some tweaking at the edge cases or something right and one of the things that's funny in equity the idea is you have to bring in someone a judge or someone who can come in and say what did they mean when they were coming up with the original law not what does the original word say but what do they mean to do the substance of it right and similarly here what i guess i'm arguing for is the social aspects of the mind probably don't need to be divorced from the model entirely the hand-waving aspects of the mind the metaphysical aspects of the mind because they are they i think are always the foundation of the possibility of a mind right we come into social you you've come in life feral and then you are and then you learn that there's a society in other people and you have self another you learn that as a child now maybe you would learn it out in the wild maybe not i guess what i'm saying though is we the model it feels like doesn't ultimately have to be mathematically 100 percent rigorous because the borrowings in the model that are made from society intrinsically have caused leakage into into that model being affected by the externality so i guess both the perception and the other model contents feel like they're unavoidable anyway just a few ruminations there and and do with do with what you should with that with the cud that i just served up i mean i think this what you're speaking to here is a pretty um big topic in the active inference community in general and this idea that you know the map is not the territory um and depending on who's speaking we kind of get lost in that a lot of the times like are we agents with generative models or are agents generative models and to what extent is this a description and to what extent is you know that's i'd rather not wait in there to be honest no i get it and i like your statement the math is not the territory it's nice because really that's nice where you think the math is the tool to understand journey or whatever but that's a nice i like that statement you just made about the math is not the territory because it often feels like and that may be something that is helps with the invitation you know the space can be threatening to non-math people because it does need a rigor i get it but there are things that are being said by the math that need to be conveyed to non-math people you know they're important things when i first dropped into the space i said oh my god all the contracts i ever made are synthetic markov blankets they are i'm creating a there there that wasn't there it's not an observation of a biological system that's natural it's an artificial of creation of a thing a place where these rules apply for internal and external operation right it's it's an artifice so and we do that a lot in the world but so that was my first kind of impression

but i don't i'm never going to understand the math enough to have the rigor to be able to delve into that and understand the implications of active inference math for the law it just won't be available to me so if and to the extent that without going to metaphysics we're able to create narratives that are honest to the math realistic and also available to the non-math people that's going to be something that will lead to a lot more both understanding and misunderstanding i get it but but a lot more awareness and adoption of the models i think which will lead to more um i think better systems better policy because there's a lot to be learned from this and it's the same thing as an engineer knowing how to make a nuclear power plant work and then other people down the line knowing how to sell power but they don't know how to make the new nuclear power plant work right you need you need all steps of the supply chain of this to be delivered as a social good i guess is what i'm saying so that's a big i'm putting it all on your shoulders you guys yeah but i mean that's the hope here i mean that's i think what's at stake a lot of the time is that you know i i think about this stuff in terms of a practice in a practice content sharing and communicating on how how best to you know improve our mental well-being given given what the the insights that this might afford and given the the structure in terms of communication that it might allow it's already impacted the way that i teach it's already impacted the way that i practice personally um but then there's a whole field of computational psychiatry as well which is which is very closely related to this i mean the kinds of well-being i'm talking about or the kinds of dysfunctions that the computational psychiatry people are talking about so there's there's a lot of hope to that and it doesn't take somebody to understand um you know how to do a gradient descent on free energy to be able to benefit from that definitely typo but this is a slide we had from the dot zero just like there's so many directions that this area verges towards just how we even make sense of this field where is active inference going to connect the the pieces just i don't know any of these elements on the list that was interesting to you and give a thought on or anyone else could just raise their hand oh the hype the hyper scanning piece is really fascinating because there's a there's a obviously a deep tradition of you know these kinds of things being learned in community and in interaction with other people um and that is not obvious how that how that happens and i'm looking forward to you know that whole space developing a little bit and being able to model you know what it'd be like to learn you know what's the difference between learning through an app and learning from a teacher cool very interesting so um i see scott's hand and then anyone else so the other place is just in in the crafts world of markets you know you have information differentials are differentials and the markets are driven by

information differentials i've asserted for a number of years that cardo's equation and thermodynamics the hot and cold differential is necessary for a heat engine to function the that that is equivalent to the differentials in information differentials and markets you don't have market action if everyone has the same information and that's that idea that markets generate information price information availability things like that the crass part of this and that tends to make things get swept away in the world of craziness when you get things into markets because anything goes and you know it's just a big mess having said that the impact of this kind of analysis will be very appealing in markets because it does have a scale independent aspect to and so it allows for insights at different levels of different markets and meta markets and related supply chains that if everyone is doing this analysis then it leads to a certain kind of interoperability because you have a risk and risk evaluation interoperability right and so the value in markets if everyone started doing active inference style of risk analysis you'd be able to make observations about at different scales because you'd have data coming out of every scale that was generated by models that were similar right so that kind of aspiration it's what i'm looking at with the i'm working with banks now etc and i'm not talking about active inference or carl fristen but i'm starting to set that up as best i can in the early stages of it because it feels just like a natural thing we are biological systems our markets and our institutions are ultimately doing reproductive and nutritional opportunities at large scales you know and so there's a lot of biology that's still bound up in what we do as organisms in our scaled structures it's exciting to watch because this thing is going to come out of left field and it's like quantum computing everyone thought there was a thing in the lab for a lot of years and now people are like oh geez here we go that's going to be a big deal in other things like business and governments and and i mean you just imagine legislative activity informed by a active inference style of model where you really do probe and we get feedback right what if we had feedback from legislation every year saying this is good or not good more this less of that right i mean that's the kind of thing there's been aspirations that for years putting lobbying aside that would lead to a great efficiency in things like legislative activity and enforcement activity things like that so i'm very excited that these things are going to get taken in the directions where they weren't intended and i think it behooves the people who really understand it early on to set up some nice chunky solid notions and paradigms so when they're adopted they don't drift too far from where they should be right because the things get picked up and taken in all sorts of wacky directions out in the world so it's kind of that's what i'm talking about simplifying in a way that still does justice to the to the what's going on there and

as a person teaching meditation that's exactly the same kind of mindset you need going into these crazy markets right is something of common intention because the markets have different agendas and intention if we want the active inference intention to be carried forward we need to make sure that we have robust presentation of the model so when it's picked up that model can withstand the abuses that happen out there in the rhetorical markets anyway again more just load on your shoulders any thoughts there blue or dean or dave i didn't mean to bring things to a screaming halt sorry i can go in a different direction no all good um um what's a um figure or a word that you might like to go to lars or when you're communicating this to audiences like more from the maybe non-active infant side what is your entry point is it one of the key words here or one of the figures here or how do you enter do you show the action perception loop do you just go straight to the bayesian graphical model um yeah good question um i think this this this card is a particular one but um i think i when i talk about this paper i kind of do it in in two steps starting with figure um well when we make when we look at precision so early on in in just talking about attention so yeah exactly here just starting here because it's quite straightforward as a model of of and attention ultimately and you can kind of see here um intuitively what this precision is about and and we can yeah let's do this let's kind of build it up and we'll go straight to it we'll go up to figure 12 which is where i'm really excited about and going in the future which is this kind of generalization of this this across and what it might mean and the game that we have to play here a little bit is a little bit dicey in that in that we're trying to make intuitive phenomenological interpretations of maths right that's not not easy and requires a lot of discussion back and forth but that's why we started here with attention because it's the one place where it's less controversial and more widely sort of defined and so we're starting with the tension here just the precision on you know the likelihood matrix and you can kind of see just intuitively i mean maybe this depending on who you are listening to this is like that's already kind of obvious yeah um you know higher precision on the likelihood matrix means you know uh is related to attention but just to kind of motivate that a little bit more subjectively um can tell that um at the center of your vision you know as opposed to the periphery you're a bit more confident about what you're seeing like if you see a glass on the table at the periphery of your vision you might not be sure if it's a glass or a mug right but you can still see it and then when you look over to it um then you can be more confident in the causes of your sensory observations right or in this case just lost lars for a second but just to sort of while he's reconnecting that's one sort of experiential urine degenerative model type thought experiment or really commands for covert

action for attention which is to realize that the color detection outside of the center of vision as well as the resolution is very low and that there's a significant blind spot so there's three features of your visual field for healthy vision that make it so that the visual field is not just seen homogeneously like a sort of silicone light detector light detector device in the camera so just a starting point for saying okay something else is happening such that attention isn't being paid it's being normalized in some way not just increase the resolution outside of the high resolution detection zone and colorize it in many cases but also to normalize the absence and paste over the blind spot so then that can be extended to other sensory observations what do you mean by normalize the absence and pain over the blind spot that part like it feels normal wha was being perceived under just again this is the whole interesting question about what is perceived as normal at what scale oh it's weird there's a door here how weird what about the door i don't want to be so i don't want to be normative with the way that i'm talking about the mental states but there's different ways that that uh anomaly can be detected and there's certain sensory illusions that are really illusions because the alternate perception which turns out to be factually different than the actual sensory o that is different but that difference structurally or inaccuracy or around resolution is just seen as part of not salience but not salient part of the organism's experience phenomenologically not to say unimportant at a nested level of analysis hey lovers welcome back we're just talking about um generative model vision so continue on too perfect thanks thank you no hopefully it doesn't drop out again um so that that's kind of motivating the phenomenological translation let's say and then and then the next step is to then say well okay we are somehow in control of that precision and here i leaned um on existing work by jakob blimenoski and fristen and others whereby that notion of control and mental action then you can you can talk about that as a deployment of precision it's some kind of a a precision um modulation is the word i'm looking for and so then the natural the kind of next step there is to make that possible within the active inference framework in the same way the other action is also possible and that's where the second level comes from um in figure four i think it is um because then this is what then allows that to be the case and really all that were all that we've done here is to say okay attention is this kind of thing this is this kind of precision um and we can motivate that in all kinds of different ways both sort of computationally and intuitively and we know we can control it so what would it mean to be able to control it well the same thing that it means to control anything within this framework and then here you go this is what kind of makes that makes that possible um and again luckily i'm really standing on the shoulders of giants here and in that you know a lot of this stuff was

already worked out you know casper italian effective inference already had this in place they just didn't have a policy policy set over this higher level yet um and so this is cool but uh like i said in the beginning uh the the real aim of the game or what really allows for um attentional freedom in in the kind of practice context is not so much um uh the attention itself right but it's your awareness of where the attention is so then um the question for me was well how do we pay attention to attention but once we have this this this structure well then that becomes kind of straightforward and not straightforward i mean you have to make a little bit of a leap but then you say well we pay attention to it in the same way that we pay attention to the sensory states and that's where you get the third level um and where these states now are the degree to which you are aware of your attention um and you know if i just ask you know how where are you of where your attention is right now um when i asked the question probably not so where just afterwards probably a little bit more aware and so there was a transition there that that happened in that higher level state which is then modulating the precision of the lower levels now this kind of uh this kind of a move computationally mathematically internally coherent and allowed but it's also not unique there's no reason why you can't do the same treatment to other precisions in the model the other ones that we're most familiar with from the literature are the precision on g model precision and the precision on b which are usually which is usually related to sort of volatility beliefs sort of um unexpected uncertainty i think if i'm saying that right um and um just as a side note we made the decision to kind of change the notation here uh just because we're talking about so many different precisions in the literature up until here precision on a was zeta i think a precision on b was omega whereas here now we've just called all precision as gamma and with a subscript based on which which parameter they're referring to gamma previously was was really just the precision on g so in case that's causing any confusion the the idea then uh and what i think is exciting is to then think about all right well if we do the same treatment right what does it mean to have this higher level state now s_1 s_2 as superscript 2 which parameterizes other precisions within the model because that now goes from an attentional state to something quite so we can talk about these individually from a practice perspective this is what this is what i'm most enthusiastic about because there's a lot more to meditation practice for instance and also just sort of cognitive mental or catalog of mental action than you know there's lots of other things that we're doing so for example the notion of acceptance or equanimity um would be the technical term where does that live what was it what does it mean to be in a state of right and here we could uh define that as a as an s as a factor of s_2 stay which is modulating the precision over our

preferences gamma c to kind of give an idea here of how we can start talking about this so a state of acceptance would be you know or state of sensitivity would be the opposite is to what degree are we allowing our preferences to impact our actions effectively uh and so you know in the in the example uh that you said in the beginning there dean of like sitting still you know how important is that i think yeah that came from you then a big part of that is acceptance right is being able to be okay with the discomfort that's arising from you know whatever whatever is going on and that's also something that we're explicitly teaching that is a that is a instruct that is an instruction that is important and a big part of what makes it all work and when i say it all work it's leading to happiness right like it's leading to well-being um and so you can start to you can start to do that game with these different different preferences you know if you outside these different precisions and see what it would mean to affect a policy over them and then you have to do this like i said kind of um delicate work of translating that into you know a phenomenological kind of uh interpretation but i don't think that's impossible and i think a lot of it lends itself quite well intuitively to be able to do that because of this sort of bayesian um space that we're in of beliefs and beliefs of beliefs and different kind of forms and the way that we talk about precision and sorry preferences and habits the point there for me is that you can start to figure out what is causing or what a kind of instruction you would need to alleviate different kind of um dysfunction or um unpleasantness so for example ryan smith um put out and colleagues put out a great paper recently on gut inference um and the point there is they're trying develop a model for you know the ways in which inferences from the gut could go wrong and try to see if that could be related to overeating and one of the one of the things that's come out there is that one of the ways in which um people's you know well-being is effective comes from this tendency for priors to get frozen or like not be updated very well in in that context it means that you know people are effectively i mean the reason for that is that you're you're selecting attention away from uncomfortable sensations of hunger so if you are continuously selecting away because you have a preference against hunger sensations then over time the priors around hunger and your behavior and hunger gets frozen because you're not updating updating those with any new observations so what's the solution there right like how can we actually help somebody in that situation and well this this gives us a bit of a hint is that maybe what needs to happen there is a training in mindful acceptance of sensations of hunger because that's where in this sort of precision dynamic and cocktail of interactions things are going wrong and if we can kind of translate practice instructions or internal instructions into into this this framework then it gives it gives uh avenues for

alleviating different forms of um different ways in which things can go wrong here thanks just one comment thanks just one comment then scott and then could you mute lars speaker set up for different speakers set up for you yep you said about instructions that might alleviate some non-preferable state and that really recalls our discussions on instructionism and interactionism and how active inference lets us think one extreme with the instructions like the cues and then on the other extreme who knows how many ways it goes with what it looks like to interact and co-develop based upon a total model or even parts of a total model rather than to up high level x in blood that means you need to take the x blocker or if it's too low you take the supplement like that could be approached here oh parameters too high lower it it's that sort of biomarker driven approach that maybe we can move beyond when we have a rich generative model and then we can underlie go into underlying factors informed by information rather than just trying to follow up on mathematical parameters and um constrain people that way um if you want to say anything or scott okay scott so um that last part in the equanimity brought me back to ecwid the equity thing again i was reading from this book the aristotle thing before and it's it's interesting because in a way it's a correction to too much um sensitivity i guess and you know or the correction of the general rules and aristotle's words and it got me thinking and and this is something we may have talked about i'm just trying to remember the phrasing it feels like what we're talking about there is that the models themselves are engaging in a peer-to-peer discussion with each other to refine each other right so the models it's never one model it's right i always tell people if i'm trying to explain active inference i say if you're in the woods and you don't have a flashlight and it's dark you're not gonna run full speed ahead you're gonna stick your hand out that's the active part and feel if there's a tree and if there's a tree you're not going to run forward straight into the tree you so you now you change your model there's a tree in front of me that's how i maybe that's a bad way to explain it but that's how i explain but this the thing is it's not just a tree there's also a model of the rain and there's a wolf chasing you and there's and you and you're hungry there's 17 things going on what it feels like is in intention we bring our models to bear in a peer-to-peer discussion internally and do we do we already use active inference to model the internal model discussion among models has that already been something we've talked about in earlier session i don't remember but we know we have a zillion different things going on that are agenda in our heads that motivate behavior right and so when we're trying to correct behavior or address behavior or become aware of we bring the other models to bear to let us do that now again they're all borrowed ultimately from the externality but once they're on board can

active inference or does active incidence already has it been brought to bear to that internal discussion among the models that then informs the decision making thanks i believe the closest that we've seen one example is this nesting which we'll get back to when we return but it was mark miller at all with their discussion of happiness with the drives for like low thirst you know low overwhelmingness for one stimuli versus another there's like a sort of domain-specific optimization and then there's this cross-domain optimization that has to do with this like eudemonia well-being component whereas the sort of hedonic drive is the domain specific that's the department of thirst department of hunger and then those are being optimized in potentially recursed levels as well so that's one way it's gone was that mapping onto classical philosophical concepts well-being success and then um another direction has been also linking it up a little bit more graphically i think in this type of work but nice insights scott just one on that just a response to that when the imp of the perverse which is that edgar allan poe notion that people act against their own self-interest to show they have power it's called the imp of the perverse and the idea is that people don't take vaccines because they want to show that they have efficacy or whatever so the it's interesting what you just said explains that nicely because the state that they're seeking is a state of perceived efficacy and it may and it is overwhelming perhaps some other things like the health thing or whatever so that's kind of an interesting notion we just raised all our today is just that it we always got to keep that realism instrumentalism she said there's 17 things going on well not if you didn't model them that's a claim about how it quote is out there and that's a second order question so if it's if it's an allusion to some causal process inside of some people's head that's different than saying i fit a model that has this hidden variable and that is this intersection of mathematics and qualitative experience is so tantalizing and so important to how we think about it and frame it so it's nice insights though blue so what scott just said made me think of um you know scott scott was talking about how our own models interact with one another right like the desire for thirst or hunger or you know to regulate temperature or whatever um but but really like where can we start to model each other's models because i think that that in this like collective kind of dynamic and that was something that i brought up in the dot zero like this collective phenomenon like my model of daniel updates every time i see daniel he is as i expect him to be but then you know suddenly like if he were to be you know purple or something then i would have like the mismatch right so our models of each other are constantly interacting and and where where can we start to maybe look at this or think of this in what aspect i love that point what break what comes to mind for me is is in this context of as you are i mean to

respond to both of your points really like the what we're talking about here is a model of our model right as you move up in the hierarchy you are modeling your own model you're getting some beliefs over how the precision in the lower levels are evolving and so you've got a model of your your model as you go up but what you're also learning there is how your mind works and how you know different states lead to different like affect for instance and as you continue to do that right as you continue to learn how your own mind works and you get more and more awareness of how things go in your mind well then naturally that you you are also learning about how other people's minds work as well there's always this kind of tension between sort of compassion in practice and awareness in practice or tension there's this kind of duality between the two and they kind of seem to come come together out of a model like this whereby you're training in awareness of your own mind and what you end up discovering is all how difficult it is actually sometimes for things to go well and fundamentally that the one thing that's driving it all is your own desire to be happy as you learn that as that really becomes part of your own model then that naturally projects onto everybody else and you kind of get this rebound of oh but then everybody else is also trying to be happy and it's hard for them too and you get this kind of natural compassionate response then that comes that comes as a result that i think also is related to this idea that scott was talking about before which is that none of these things happen in a vacuum like my own well-being is very so so intertwined with your well-being and so all these things kind of grow then nice it's almost like in in scots forest one element is there's other people and maybe non-people entities in the forest and then another aspect would be like you can modify your niche you can you can walk different ways you can use technology and it's sort of that tension between how sort of slam dunk it is for a very constrained case wanting and sort of hinting that it could generalize that more because of the ease at which it can integrate not just within a domain like the mountain car or just thirst but across domains especially in a way where all the benefits of the first level relatively speaking like the computability and the interpretability of parameters if one believes that then it exists also at the second level and higher levels so like figure five you have the three layers or six but then twelve the generalized model like you only had two with it i guess implicit hint that you just do it again yeah yeah it's just yeah so pretty cool dean d i mean just to just to speak to that point i mean um you you can and i have um and you can start to think about you know the the third the sort of third level equivalent of attention was sort of awareness of mental states right so now it's like you're not you're not just aware of where your attention is but you're aware of how much you you know to what extent do i want here to what extent am i

certain about my environment to what extent am i happy to what extent am i um doing things automatically to what extent am i being influenced by my memories and priors right that to go through the whole gambit there but then you could do the same thing with b at the third level gamma b that's okay so what is what is that state telling and so well to what extent is my mind volatile so how confident am i about the transitions of my mental states right and let's say you take a psychedelic or something and all of a sudden your mental states are all over the place well then maybe that precision at the third level would drop because now you don't have the ability to predict how your mental dynamics are moving or see you know to what extent am i am i wanting or not wanting particular mental states um as opposed to sensory observations you know like how much how much attachment do i have to joy and fear right now um is what you'd be inferring at that level and you can kind of keep keep going keep going through that thanks yeah so i know we're getting close to the end here and i just kind of wanted to bring this up so my history just a little bit of context was that i i was a context generating programmer for uh high school kids and i wanted to take them from a place of subliminal to liminal to understanding what the edges were what the limits were so my my basic job was unnesting learners having them transition but having them take a model outline along with them so they weren't kind of going naked into the forest so what one of the things that we used to talk about in this transitioning across basically the deep generational transformation was using a street metaphor starting with perceiving so which is which kind of is unplanned and moving to thinking about thinking which is plan and attend and attend and plan and then this was the critical piece realizing the inside the lines versus the outside the lines and with the example that we use is that there's a kind of a crosswalk logic calculation problem so you use logic to push the button that it gets the flasher going before you enter the crosswalk you stay inside the lines you even have perfect gating and balance so that you can look at your phone while you're crossing the street through those lines of immortality and now you're dead because there was a car a driver driver in a car who was also looking at their phone and then an accident occurred so the question you had then was did you follow the logic did you use the rules and the answer was yes and were you aware of the free energy gap and the answer was well not anymore because logic tends to point at the jaywalker and ask what are they thinking i mean come on they're taking the risk here they're prob those jaywalkers are probably thinking about the opportunity that there's no cars coming at me right now or on the street they're not present so they're thinking about the context and the directionality of the transition and they're also thinking about the rate of travel how fast can i get across the road and will i

get a ticket will there be somebody who's going to catch me in and pull me over all together who's making a decision based on more information from which to update and i think this is the big question around precision after all our parallel lines in the street and in the mathematical operations context supposed to represent logic and precision i think they are and in this case and based on this slide right here this is why i liked you talking about math is not the territory math does allow for the sketching out of territory and sometimes we can get inside the lines and think that we are using the logic when in fact we're actually going to be more precise and actually use what i would think to be logic based on more information once we get outside those lines so when i think when we get into the point two of this i think that's going to be a really interesting point because steven talked about it was steve and you were wondering lars who was the person who's talking about being really still that was stephen before he had to take off i wanted i'm kind of wanting to think about this as being really active and being really contemplative and being really precise and then be able to sort of tie that in with the um the metacognition piece because i think it's the transition here that all of us are trying to parallel or trying to sort of piggyback on your paper and i think it's i think it's i think that's the real value in this conversation i also think that we can get outside the lines and learn more than we often do while we're inside the lines because you know i'm immortal now i've got these two white lines protecting me and that's not always necessarily the case yes yes exactly i mean this what you're talking about there for me speaks to this power of suspended belief or uncertainty in our own maintaining a sense of uncertainty or in our own model so that no single belief can become so rigid and enshrined that it's resistant to updating in the future and if you spoke to inesh and mark miller recently you know there there is kind of down the same line that a lot of a lot of dysfunction a lot of um low states of well-being comes from that rigidness and that freezing of of beliefs that then you know leads all kinds of afferent behavior um afterwards because you're trying to fulfill the expectations of belief that isn't being updated the the what's on offer here is a mechanism to describe how you might remedy that solution by deliberately controlling uh or inquiring about the the metacognitive belief so gamma b in this case uh over my priors and if i can control that state transition well then it gives me the opportunity to relearn that b matrix at the bottom because i've injected a little bit of uncertainty and now i'm more able to sample information that is counter to what i know and allows for you know just better optimization ultimately yeah take the model along thanks dean and awesome insight there so just in the last couple minutes to dave any textual questions live chat any final questions and then scott and then blue if you'd like to make a last comment so

go ahead scott so that um dean what you were just saying there goes to that what i was trying to fuss with on that equity point before where it's the rules of general application and what i realized while you were saying the reason the equity is so fascinating to me is what they did in law and equity is they said okay the rules of general application don't always lead to the right result and what they've now done is collected all the remedies for the bad results and can characterize those into groups so you may have a rule of general application and then there's something out there in equity called unclean hands which means if i did something wrong if i'm a burglar i can't sue you if i trip over your coffee table right that's unclean hands right or the doctrine of um unjust enrichment means if i find some money and or if i get some extra money you pay me too much give me an extra 20 bill i say i keep it because you you gave it to me but it wasn't meant to be it's unjust enrichment so equitably you should give it back where i find someone's wallet all this stuff right so the reason i when you were just talking made me realize you know it may be that we apply the rules of general application and then we find some exceptions and then when we start or some ways they don't apply the right way in active inference and maybe we start to cluster those bad results let's call them or unintended results or something start to cluster them and characterize those like they did in equity so that's what i was groping for before is the rules of general application can have great value and they need those corrections and those corrections themselves may have clusterings and groupings that themselves can be managed thanks great discussion by the way thank you so much lars is fascinating stuff yep thanks scott blue yeah i'm looking forward to more next week hopefully if you're available um and maybe kind of probing the different types of meditation and i'm not sure how familiar you are with different kinds of meditation but i've done a bunch and highly variable and maybe whether or not they all fit this model or not is something i've been kind of thinking about and i'm gonna think a little bit more about that um over the next week so thanks for the great discussion thanks blue i think it will be good to see if anybody who's listening in the intervening week wants to join or if they just want to watch along next week so they can ask questions and uh i think exploring that aspect of group active inference agents by design or out there in nature that we've been discussing a lot you know when is it is it can we model it that way and then asking well can't we have nested metacognitive models of groups what does that mean is it a different philosophical bridge is it two exits on the same freeway what is going to be the relationship great times thanks lars and if you have any final comments otherwise we'll end no thank you this has been really great i had a really fun conversation thank you for your kind of openness and great questions and i just really enjoyed the energy

here it's not been a at all if you see what i'm saying so thanks awesome peace out everybody
see you next week